

Kongkun Chen

301-273-4175 | kchen137@umd.edu | [linkedin.com/in/kavi-chen/](https://www.linkedin.com/in/kavi-chen/) | github.com/Kaitou0309

EDUCATION

University of Maryland, College Park

Bachelor of Science in Applied Mathematics, Computer Science - Machine Learning

MD

Aug 2026

- **Honors and Awards:** Meritorious Awards in SCUDEM, 2022 - 2026 Dean's List
- **Relevant Courses:** Natural Language Processing, Machine Learning, Algorithm, Data Science, Database Design, Applied Probability & Statistics

EXPERIENCE

Machine Learning Research Intern

Earth System Science Interdisciplinary Center (ESSIC)

Jun 2025 – Present

College Park, MD

- Built scalable **TensorFlow data pipelines** for large satellite datasets using **tf.data** and **TfRecord**, enabling streamed GPU training, on-the-fly augmentation, and reproducible experiments across seasonal data splits.
- Developed and benchmarked **RDN**, **RRDN**, and GAN-based super-resolution models for **brightness temperature satellite data**, studying architectural depth, residual connections, and physically consistent upsampling for $4\times$ reconstruction.
- Designed and compared **custom loss functions**, including **MAE**, **SSIM-based composite loss**, and **physics-informed gradient loss**, to balance numerical accuracy, structural fidelity, and sharp edge reconstruction in atmospheric fields.
- Built an evaluation and ablation workflow using **RMSE**, **PSNR**, **SSIM**, **bias**, **latency**, per-scene diagnostics, and hard-scene analysis to identify failure modes and guide model and loss-function iteration.

PROJECTS

CrIS Temperature Retrieval | *Python, TensorFlow, Keras, Scikit-Learn, NumPy, Matplotlib* Jun 2025 – Aug 2025

- Built a regression pipeline to retrieve **atmospheric temperature profiles** from **CrIS hyperspectral infrared satellite data**.
- Compressed highly correlated spectral inputs with **PCA** to the top **40 principal components**, then compared **ANN**, **CNN**, and **ResNet** models, scaling the final architecture to **8 ResNet blocks**.
- Designed a custom loss, **MAE + $0.5\times$ Gradient Loss**, achieved $R^2 = 0.78$, and used residual analysis to identify higher-variance regions near the tropopause and stratopause.

Transformer Encoder & Character-Level Language Model | *Python, PyTorch, NumPy*

May 2026

- Implemented a **Transformer encoder from scratch** with positional embeddings, self-attention, residual connections, and feed-forward layers.
- Built a causal **character-level language model** on text8 with shifted targets and masked attention, achieving **4.65 dev perplexity**.

PCA, Softmax Regression & Neural Networks | *Python, NumPy, SciPy, Scikit-Learn*

Apr 2026

- Implemented **PCA from scratch** for dimensionality reduction, explained variance analysis, eigendigit/eigenface visualization, and reconstruction.
- Built **softmax regression** and a modular **feedforward neural network**, reaching **92.18%** and **90.93%** MNIST test accuracy.

TECHNICAL SKILLS

ML / Frameworks: TensorFlow, Keras, Scikit-learn, Numpy, SciPy, Pandas

Programming: Python, Java, Rust, C/C++, MATLAB, OCaml

Databases: MySQL, NoSQL, MongoDB

Data / Visualization: Matplotlib, Seaborn, TfRecord

Tools / Systems: Linux, Git, SLURM, HPC, VS Code

Certification: Society of Actuaries – Exam P (Probability), Passed July 2025